6 Results, validation and discussion

We now report the comparisons of Table 2; displacements refer to the points marked in Fig. 4.

Physical response. Under the square's load, the loaded face of the plate with a hole (Fig. 5) moves 0.676 mm instead of 0.492 mm: removing material makes the plate softer. The load must flow around the hole, so stress gathers at its top and bottom. The largest element stress σ_{xx} there rises from $2.80p$ to $3.68p$ over the four meshes and has not yet settled. At the inner corner of the bracket the displacement settles (0.124, 0.127, 0.128, 0.128 mm), but the largest stress in the adjacent triangles keeps growing (119, 138, 154, 178 MPa). This is expected, since the exact stress at an inward-pointing corner is infinite [6].

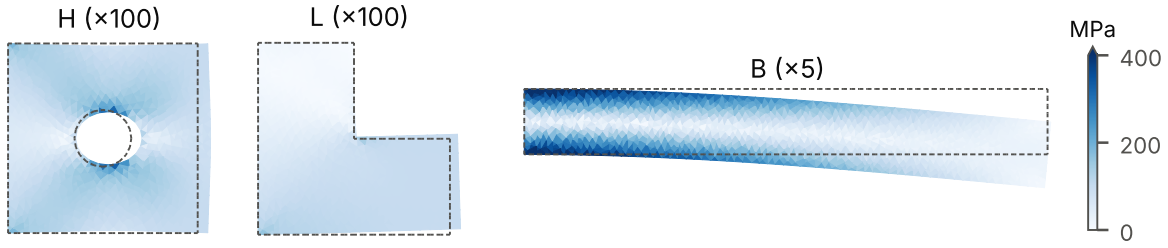


Figure 5: Deformed objects (second mesh level) coloured by element von Mises stress; dashed: original outline. Displacements are amplified 100 times (H, L) and 5 times (B).

The same 0.5 MN applied to the upper arm of the bracket moves the loaded face 0.304 instead of 0.472 mm and halves the corner stress (71 instead of 138 MPa), but the peak stress moves to the clamp and reaches 332 MPa. The beam is far more flexible in bending than in stretching: its tip moves 50.8 mm under a 50 kN transverse load but only 2.00 mm under a 500 kN axial load. A roller and pin instead of the clamp on H raise the displacement by only 0.9% but $\kappa_2(A)$ by 38%: fewer held components make the softest mode softer.

Displacements scale exactly with the load ($p = 25$ to 250 MPa) and with $1/E$ ($E = 70$ GPa), while stresses and $\kappa_2(A)$ are unchanged, since E only scales A. Poisson's ratio changes the shape of D: raising ν from 0 to 0.45 lowers the displacement by 1.7% but raises $\kappa_2(A)$ by 37% and the CG iterations from 278 to 332.

Mesh refinement and discretization error. Under refinement the matched-point displacements settle. For the square the change between levels falls from 4.3×10^{-4} to 5.0×10^{-5} mm, and at the loaded-face midpoint the finest irregular and structured meshes agree to 2×10^{-4} (relative). Near the hole, where stress varies rapidly, convergence is slower (changes from 2.3×10^{-2} to 2.2×10^{-3} mm). The beam's tip deflection, 48.3, 50.8, 51.3 and 51.5 mm, approaches the beam-theory value of 51.2 mm, or 51.8 mm with shear deformation [5], from below: linear triangles are too stiff in bending [1]. The finest mesh is 0.7% below the estimate with shear. These changes, at least 10^{-4} relative, measure the discretization error. The algebraic error of every accepted CG solution against sparse LU is at most 1.0×10^{-9}, so the mesh, not the solver tolerance, limits the accuracy.

Conditioning and solver performance. Figure 6a shows that $\kappa_2(A)$ grows in proportion to N for every object, with fitted log–log slopes 0.98–1.02. This is the growth like h^{-2} expected for finite element stiffness matrices, since $N \propto h^{-2}$ in two dimensions [2]. By Eq. (11), CG iterations should then grow like $\sqrt{\kappa_2} \propto N^{1/2}$; the measured slopes are 0.50–0.52 (Fig. 6b). Shape changes the constant, not the rate. The beam's κ_2 is about 260 times the square's at similar N: bending a slender beam takes little energy, so its smallest eigenvalue is 290 times smaller. Yet CG needs

only 1.8 times as many iterations, not $\sqrt{260} \approx 16$ times: only four beam eigenvalues lie below the square's smallest, and CG removes a few isolated eigenvalues in a few extra steps [3].

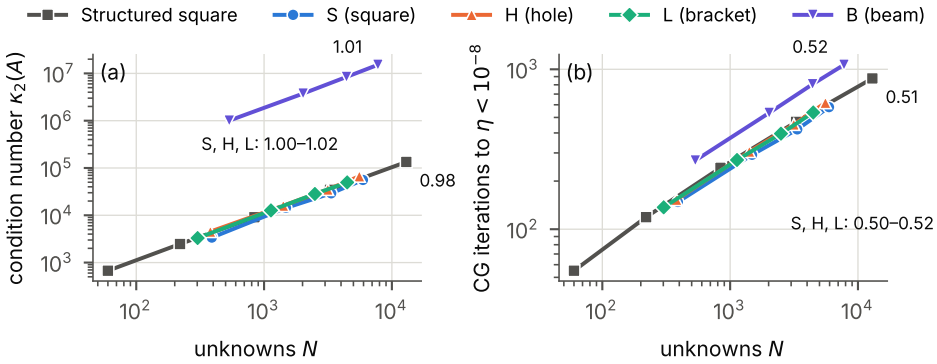


Figure 6: (a) $\kappa_2(A)$ and (b) CG iterations against N, with fitted log–log slopes; the structured square is that of Section 4.

Table 3: Solvers on the second mesh level: iterations / best time in ms. *: $\eta < 10^{-8}$ not met.

	S, $N = 1484$	H, $N = 1414$	L, $N = 1132$	B, $N = 2020$
$\kappa_2(A)$	1.43×10^4	1.60×10^4	1.27×10^4	3.75×10^6
Sparse LU	4.6	4.6	2.9	5.0
CG	292 / 10.8	306 / 11.1	271 / 6.9	534 / 24.0
Diagonal PCG	264 / 10.6	281 / 12.2	246 / 9.5	493 / 20.9
MINRES	285 / 26.0	302 / 26.7	268 / 20.5	541* / 50.5
Gauss–Seidel	20,000* / 11,700	20,000* / 11,700	20,000* / 10,600	20,000* / 14,800

In Table 3, diagonal PCG saves only 4–12% of the CG iterations and no time: all triangles have similar size and the material is uniform, so the diagonal of A varies little and scaling by it lowers κ_2 by only 12–31%. Jacobi diverges on every mesh, as ρ_J of Eq. (10) is 1.09–1.44. Gauss–Seidel reaches the target only on the coarsest meshes of S, H and L; on the beam it is still at $\eta \approx 0.2$ after 20,000 sweeps. Sparse LU is the fastest method on every system, including the largest ($N = 7776$: 31 ms against 162 ms for CG).

MINRES fails the test on all four beam meshes (final η from 1.6×10^{-8} to 1.5×10^{-6}), yet differs from LU by at most 6×10^{-11}. In floating point, the residual of a very ill-conditioned system can stall at a level that grows with κ_2 times the machine precision, and the internal residual estimate of MINRES can drift from the true one [4]. MINRES thus stopped above the target: a failure under our protocol, though the answer is accurate.

Validation and sensitivity. Prescribed displacements are exactly zero in every case. Applied loads and the reactions of Eq. (7) balance in force and moment to a relative error below 5×10^{-14} for the plates and 3×10^{-10} for the worse-conditioned beam. Every accepted iterative solution agrees with LU to 1.4×10^{-7} or better. Held only horizontally, H can slide vertically: then $\lambda_{\min}(A)/\lambda_{\max}(A) \approx 10^{-16}$ and Cholesky fails, whereas every supported case has $\lambda_{\min} > 0$. A random load perturbation of relative size 10^{-6} on the 20×20 structured square ($\kappa_2 = 9104$) changed the displacement by 4.5×10^{-7}, far inside the bound 9.1×10^{-3} of Eq. (13).

Limitations. We tested only diagonal preconditioning, not incomplete Cholesky or multigrid. Element stresses are constant per triangle, so peaks depend on the mesh. The beam (up to 540 MPa) and the 250 MPa case exceed the yield stress of steel, which the linear model ignores. Timings come from one machine and implementation; only ratios and trends are meaningful.

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